
Sensor-to-Language Alerts and Voice Queries for Smart Collar MWD Health Monitoring

An NLP Framework for Interpretable Health Monitoring of Military Working Dogs

Project Presentation

March 2026

Abstract

This project proposes an NLP-based framework to translate raw sensor streams from MWD smart collars into human-readable alerts and voice-query responses.

- **Problem:** Existing systems output raw vital signs that are difficult to interpret under field pressure.
- **Proposed Solution:** Integration of anomaly detection with Transformer-based Natural Language Generation (NLG).
- **Innovation:** A multimodal approach combining physiological sensors with speech interfaces (OpenAI Whisper).
- **Impact:** Reduces cognitive load for handlers and accelerates emergency response times.

Introduction & Problem Statement

Raw sensor data is unintelligible to handlers in high-stress operational environments.

- **Operational Stress:** MWDs face extreme heat and exertion during missions (Foster et al., 2022).
- **Cognitive Load:** Handlers must manually interpret numerical vitals while managing combat tasks (Mealin et al., 2015).
- **Information Gap:** Systems lack semantic context for "why" a reading is high (Boursalie et al., 2018).
- **Interpretability:** Current alerts are binary thresholds, not descriptive insights (Zhang et al., 2025).

Based on a review of 20+ foundational papers identifying the gap between sensing and action.

Related Work: Sensors & NLG

Wearable Data Mining

- Anomaly detection techniques for physiological signals (Banaee et al., 2013).
- Feature-based analysis of heart rate and respiration (Boursalie et al., 2018).

NLG in Healthcare

- *BabyTalk* system summarizing neonatal ICU streams (Portet et al., 2012).
- NLG for prehospital care reporting (Wilson et al., 2013).

Sensor-Language Models

- *SensorLLM*: Aligning motion sensors with LLM tokens (Li et al., 2024).
- *SensorLM*: Transformer foundation for wearable captions (Zhang et al., 2025).

AI Interpretability

- Benchmarking LLMs for wearable health prediction (Han Kim et al., 2024).
- Explainable AI (XAI) for canine behaviors (Shastry et al., 2025).

Related Work: Canine Audio & Voice

Bark Classification

- Classification of dog barks using Machine Learning (Molnár et al., 2008).
- Deep learning for context and emotion detection (Gómez-Armenta et al., 2024).

Behavioral Analysis

- Valence-arousal modeling for pet vocalization (Huang & Situ, 2025).
- Fully automatic voice analysis systems for bioacoustics (Karaaslan et al., 2024).

Voice Interfaces

- Audio monitoring for blind handlers (Mealin et al., 2015).
- Speech-to-text (ASR) performance on edge devices.

Retrieval Models

- RAG models for knowledge-intensive NLP (Lewis et al., 2020).
- Retrieval-augmented chatbots for clinical advice.

Gaps Identified

1

No Interpretability

Existing collars output raw numbers or simple beeps; they lack textual explanations of dog conditions.

2

Lack of NLG in Animal Health

Prior sensor-to-text work is focused on human healthcare (NICU); there is a gap in animal-specific vitals translation.

3

Isolated Modalities

Audio (bark) analysis and physiological monitoring are separate fields; no system integrates both in an NLP framework.

4

Edge Constraints

Large foundation models (SensorLM) require high compute; a lightweight, deployable solution for the field is missing.

Project Objectives

01

Anomaly Detection Pipeline

Design a system for MWD sensors (heart rate, temp, GPS) to detect events like tachycardia and hyperthermia.

02

NLG Module Development

Map structured sensor events to concise, human-readable status reports and actionable warnings.

03

Speech Interface Implementation

Enable hands-free handler queries (e.g., "Is Rex OK?") using on-device speech-to-text models like Whisper.

04

Full System Integration

Deploy on wearable and mobile edge devices: sensor collection
→ anomaly flagging → NLP output.

Proposed Architecture



Multimodal Fusion: Combines vitals with on-device speech-to-text for hands-free situational awareness.

Open-Source Tool Stack

HF

HuggingFace Transformers

Pre-trained T5/BART models for alert generation.

W

OpenAI Whisper

Efficient speech-to-text for hands-free queries.

LC

LangChain

Orchestration of the sensor-to-LLM pipeline.

SP

spaCy

Entity & intent recognition for voice commands.

FA

FastAPI

Serving NLP models as low-latency APIs.

TL

TensorFlow Lite

Deploying lightweight models on wearable edge.

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